Huawei Access Node JAY-ACC-13 Optical Transport Failure SOP (ALARM_9928)

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_9928

Description

This Standard Operating Procedure (SOP) is designed to troubleshoot and resolve Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node JAY-ACC-13. The failure domain occurs due to the inherent characteristics of the Optical Transport Network (OTN) protocol, which relies on the integrity of the optical signals transmitted over fiber optic cables. In the event of SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, the OTN protocol's ability to detect and correct errors is compromised, leading to the generation of the ALARM_9928 alarm code. This SOP outlines the necessary diagnostic and resolution steps to identify and remediate the root cause of the failure, ensuring the reliability and integrity of the optical transport network.

Pre-requisite Checks

- check: Verify that the Jayanagar Metro Site is accessible and that all necessary personnel and equipment are on site., actions: ['Dispatch remote hands to the Jayanagar Metro Site, if required.', 'Check the status of redundant links to ensure they are functioning properly.']

- check: Verify that the Huawei Access Node JAY-ACC-13 is powered on and in a stable state., actions: ['Check the alarm panel for any other active alarm codes.', "Verify that the node's management interface is reachable and responding to Telnet/SSH connections."]

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: display optical transmit alarm, Expected Output: ALARM_9928: Optical transmit alarm detected, Description: This command displays the current optical transmit alarm status, allowing the technician to verify the presence of the ALARM_9928 alarm code.

Command: display optical receive alarm, Expected Output: ALARM_9928: Optical receive alarm detected, Description: This command displays the current optical receive alarm status, allowing the technician to verify the presence of the ALARM_9928 alarm code.

Command: display sfp xfp info, Expected Output: SFP/XFP serial number: 1234567890, Description: This command displays the SFP/XFP
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information, including the serial number, allowing the technician to
verify the SFP/XFP's identity and check for any firmware or hardware
issues.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp xfp 1/1/1 with sfp-xfp-100g-lr, Description: This step involves replacing the faulty SFP/XFP module with a new one. The technician must ensure that the replacement module is of the correct type (100G-LR) and is properly seated in the slot.

Step 2: Step: Verify fiber connectivity, Command: display optical fiber info, Description: This step involves verifying the fiber connectivity and ensuring that the fibers are properly connected and not damaged. The technician must check the fiber's continuity and verify that the fibers are properly secured.

Step 3: Step: Reset the optical transmit and receive alarm, Command: reset optical transmit alarm, Description: This step involves resetting the optical transmit and receive alarms to clear the ALARM_9928 alarm code. The technician must verify that the alarm is cleared and that the node's status is stable.

Post-Check Verification

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Command: display optical transmit alarm, Expected Output: No alarm
detected, Description: This command verifies that the optical transmit
alarm is cleared and that the node's status is stable.

Command: display optical receive alarm, Expected Output: No alarm
detected, Description: This command verifies that the optical receive
alarm is cleared and that the node's status is stable.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp xfp 1/1/1 with original-sfp-xfp-100g-lr, Description: This step involves restoring the original SFP/XFP module that was replaced during the resolution steps. The technician must ensure that the original module is properly seated in the slot and that the node's status is stable.

- Step: Re-apply the optical transmit and receive alarm, Command: apply optical transmit alarm, Description: This step involves re-applying the optical transmit and receive alarms to restore the original alarm state. The technician must verify that the alarm is re-applied and that the node's status is stable.